

Quiz 2 Solutions:

1. $A_c = 1$, $A_m = 2$, $f_c = 100 \text{ Hz}$, $f_m = 5 \text{ Hz}$.

(a) $s(t) = A_c \cos 2\pi f_c t + A_m \cos(2\pi f_m t) \cos(2\pi f_c t)$
(USB)
 $= A_c \cos 2\pi f_c t + \frac{A_m}{2} \{ \cos(2\pi(f_m + f_c)t) + \cos(2\pi(f_c - f_m)t) \}$
(LSB)

(b) $m(t) = A_m \cos 2\pi f_m t$

$\hat{m}(t) = A_m \sin(2\pi f_m t)$... Hilbert Transform
 ~~$\hat{m}(t) = A_m \sin(2\pi f_m t)$~~

$s_{\text{USB}}(t) = m(t) \cos(2\pi f_c t) - \hat{m}(t) \sin 2\pi f_c t$

$= A_m \cos(2\pi f_m t) \cos(2\pi f_c t) - A_m \sin(2\pi f_m t) \sin(2\pi f_c t)$

$= 2 \cos(210\pi t) \dots$

$s_{\text{LSB}}(t) = m(t) \cos(2\pi f_c t) + \hat{m}(t) \sin 2\pi f_c t$

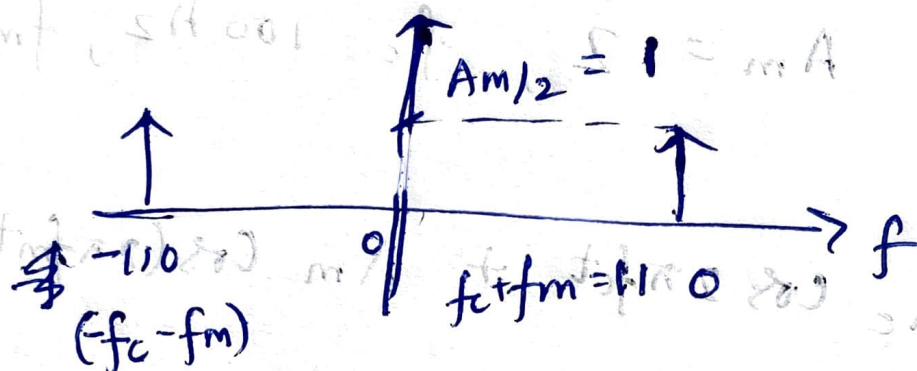
$= 2 \cos 190\pi t \dots (')$

(Don't
give any
partial
marks)

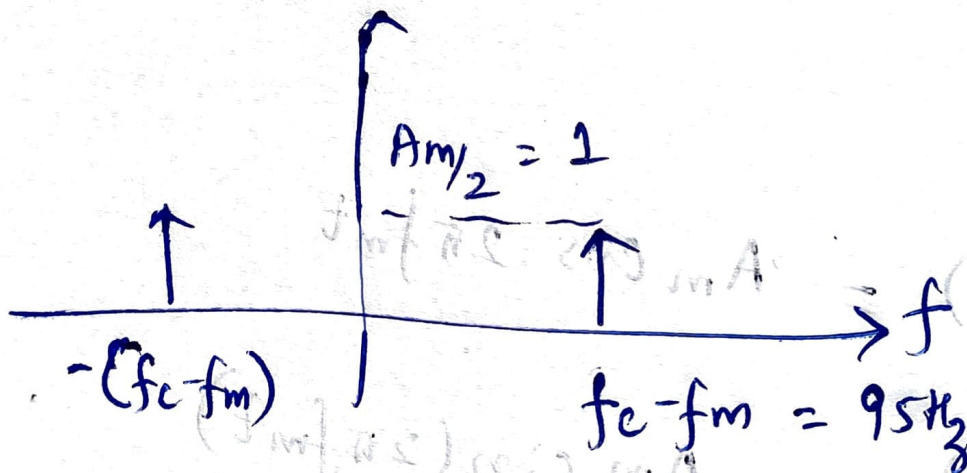
c)

$S_{USB}(f)$

No Carrier



$S_{LSB}(f)$



$$2) \quad P_t = \left[1 + \frac{\mu^2}{2} \right] P_c$$

earlier: $P_t = P_c$ for $\mu = 0$

for $\mu = 1/\sqrt{2}$ $P_t = 5/4 P_c$

$$\% \text{ change} : \left(\frac{5/4 P_c - P_c}{P_c} \right) \times 100\% = \frac{1}{4} \times 100$$

Increases by 25%

(1)

(dim (0.5))

$$(3) \quad c(t) = 10 \cos(2\pi \times 5 \cdot 10^8 t)$$

$$m(t) = 4 \cos(2\pi(1250)t)$$

PM wave:

$$v_{PM}(t) = A_c \cos(2\pi f_c t + k_p m(t))$$

$$= 10 \cos(2\pi 5 \times 10^8 t + k_p \cdot 4 \cos(2\pi(1250)t))$$

Given : $k_p = 2\pi$

Instantaneous freq : $\omega_i(t) = \omega_c + k_p m'(t)$

$m'(t) = -4 \times 2\pi \times 1250 \sin(2\pi(1250)t)$

$\omega_i(t) = 10^9 \pi - 16\pi^2(1250) \sin(2\pi(1250)t)$

Freq deviation :

$$\Delta f = \frac{k_p m'_p}{2\pi} = \frac{4 \cdot 2\pi \times 1250}{2\pi} = 8\pi \times 1250 \text{ (no units)}$$

$B = 1250 \text{ Hz}$

$\beta = \frac{\Delta f}{B} = \frac{8\pi}{(0.5)} \text{ (no units)}$

Carson's Bandwidth :

$B_{PM} = 2B(\beta + 1) = 2(1250)(8\pi + 1) \text{ Hz}$

--- (0.5)

(important)

(4) (1) FM modulated :

$$I : A_c \cos \left(2\pi k_f \int_{-\infty}^t m(\tau) d\tau \right)$$

$$Q : A_c \sin \left(2\pi k_f \int_{-\infty}^t m(\tau) d\tau \right) \quad \dots (0.75)$$

(2) PM :

$$I : A_c \cos (k_{pm} m(t))$$

$$Q : A_c \sin (k_{pm} m(t)) \quad \dots 0.5$$

Before this, they need to write the passband form for angle modulated wave:

$$v_p(t) = \left. \begin{aligned} &A_c \cos 2\pi f_c t \cos \phi(t) - A_c \sin 2\pi f_c t \sin \phi(t) \end{aligned} \right\}$$

& then write $\phi(t)$ for FM & PM

--- 0.5